

End-Semester Examination (WS: 2023-24)
GPE201 (Earth and Planetary System)

Date: 04 May 2024

Time: 08:00 – 11:00 hrs

Maximum marks: 100

General instructions: There are SIX questions, and the candidate must attempt FIVE questions in total. Question No. 1 is compulsory. The number of marks carried by a question/part is indicated against it.

Neat sketches may be drawn to illustrate answers wherever required. Parts of a question must be answered sequentially in the same location. The number of marks assigned to each question is indicated next to it. Unless otherwise stated, symbols and notations have their standard meaning. Assume suitable data if necessary, and indicate the same clearly.

Constants which may be needed:

Planck constant: $6.62607015 \times 10^{-34}$ J.s

Stefan - Boltzmann constant: $5.670374419 \times 10^{-8}$ W.m⁻².K⁻⁴

Wien's displacement constant = 2898 μ m.K

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- Q.1(a):** Explain the role of the biological pump in regulating atmospheric carbon dioxide levels.....(5 marks)
- Q.1(b):** What is the Southern Oscillation Index, and how is it utilized in forecasting El Niño, La Niña and the Indian monsoon?.....(5 marks)
- Q.1(c):** Discuss the formation of cold currents with examples, and explore their impact on marine ecosystems.....(5 marks)
- Q.1(d):** Why does the Ferrel cell, located at mid-latitudes, have westward winds near the surface but eastward winds at higher altitudes?.....(5 marks)
- Q.1(e):** If a sample of carbon dioxide in a pump occupies a volume of 25.5 mL at 50°C, what volume will it occupy when the temperature is raised to 75°C, assuming the pressure of gas remains constant?.....(5 marks)
- Q.1(f):** Define the troposphere and describe its key characteristics, including its altitude range, temperature profile, and role in weather processes.....(5 marks)
- Q.1(g):** What are the different physical properties of minerals, and how is cleavage defined in mineralogy?.....(5 marks)
- Q.1(h):** The Keeling Curve illustrates a seasonal oscillation and a long-term trend in global atmospheric CO₂ levels. Explain these two patterns and discuss their primary causes.....(5 marks)
- Q.2:** How do Ekman Transport, Ekman Suction, and Ekman Pumping collectively influence the vertical movement of water in the ocean?.....(15 marks)
- Q.3:** The planet Mars orbits the Sun at an average distance of 1.524 AU and has an orbital period of 687 Earth days. Find out (i) the orbital period of a planet C that circles the Sun at an average distance of 3.0 AU, and (ii) the additional distance (in AU) that a theoretical planet D would need to be from the Sun in order to double its orbital period compared to Mars.....(15 marks)
- Q.4:** What is the PETM? What are the most likely causes of the PETM? How does the study of the PETM help us understand potential future climate scenarios?.....(15 marks)
- Q.5:** How does silicate weathering contribute to the Earth's carbon cycle, particularly in the context of CO₂ sequestration?.....(15 marks)
- Q.6:** Two bodies, A and B, have different thermal emissivities: 0.01 for A and 0.81 for B. Despite this difference, the outer surface area of both bodies is the same, and they emit total radiant power at the same rate. The wavelengths, λ_A and λ_B , corresponding to the maximum spectral radiance in the radiation from A and B, respectively, differ by 1.00 μ m. If the temperature of body A is 5802 K, find out (a) the temperature of body B, and (b) the wavelength λ_B(15 marks)



Department of Mining Engineering Indian Institute of Technology (ISM), Dhanbad

Examination: **END SEMESTER**

B. Tech (MNE) 4th Semester

Session: **2023-24**

Subject: **MNC 205 ROCK MECHANICS**

Semester: **WINTER**

- Instructions: 1. Answer **ALL** questions
2. Neat and accurate sketches will assist you in solving the problems logically and faster
3. Assume logical data, where necessary and **state them explicitly**
4. Use the sign conventions applicable for rock mechanics.

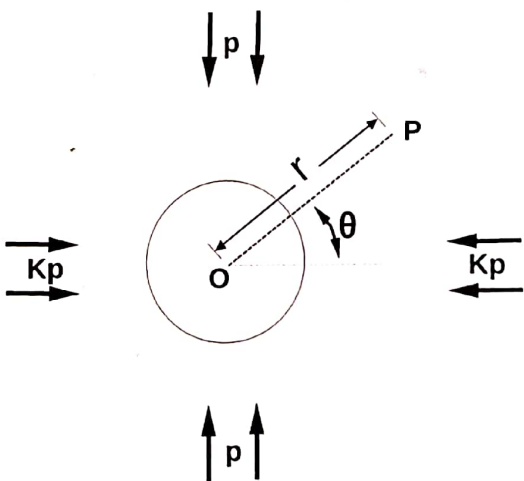
Time: **3 HOURS**

Max. Marks: **100**

Q. No.	Question	Marks																								
1	The state of stress at a point "P" is given by the following tensor $\begin{bmatrix} 25 & 10 & 12 \\ 10 & 36 & 15 \\ 12 & 15 & 28 \end{bmatrix}$ (a) Determine the magnitude and direction of the principal stresses at the point "P". (b) Determine the following on the plane $x + 2y + z = 1.6$ i. traction vector ii. magnitude of normal stress iii. magnitude of the shear stress	3 + 3×3 = (12) 3+3+2 = (8)																								
2	(a) Discuss the Coulomb failure criteria using a neat sketch of Mohr's circle. Then show that in terms of the principal stresses, the criteria can also be expressed as $\sigma_1 = \frac{2c \cos \phi}{1 - \sin \phi} + \sigma_3 \left(\frac{1 + \sin \phi}{1 - \sin \phi} \right)$ (b) Show that the slope and intercept of the best fit linear regression line on the σ_1 versus σ_3 graph can be determined using the following set of equations $\begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^j \sigma_{3i}^2 & \sum_{i=1}^n \sigma_{3i} \\ \sum_{i=1}^n \sigma_{3i} & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n \sigma_{3i} \sigma_{1i} \\ \sum_{i=1}^n \sigma_{1i} \end{pmatrix}$ (c) A set of 5-triaxial tests were conducted in laboratory. The relevant data are given as under. <table><tr><th>Specimen No.</th><th>Average diameter of the specimen (mm)</th><th>Axial load (tonnes)</th><th>Confining Pressure (MPa)</th></tr><tr><td>1</td><td>54.23</td><td>8.1</td><td>1.6</td></tr><tr><td>2</td><td>54.18</td><td>9.2</td><td>2.8</td></tr><tr><td>3</td><td>54.36</td><td>9.8</td><td>3.7</td></tr><tr><td>4</td><td>54.22</td><td>11.4</td><td>4.2</td></tr><tr><td>5</td><td>54.13</td><td>12.2</td><td>6.3</td></tr></table> Determine the effective cohesion and the angle of internal friction (assume $g = 10m/s^2$).	Specimen No.	Average diameter of the specimen (mm)	Axial load (tonnes)	Confining Pressure (MPa)	1	54.23	8.1	1.6	2	54.18	9.2	2.8	3	54.36	9.8	3.7	4	54.22	11.4	4.2	5	54.13	12.2	6.3	6 6 8
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Q. No.	Question	Marks																										
3	(a) i. Define shrinkage limit of a soil using a neat and labeled sketch. ii. A completely saturated clay has a volume of 31.25 cm^3 and mass of 60 gram. The same soil after drying has a volume of 23.92 cm^3 and a mass of 44 gram. Compute (A) the porosity of the initial sample, (B) the specific gravity of the soil grains and (C) the shrinkage limit of the soil. (b) i. Use neat and labeled diagrams to explain as to how the soil gradation curve alone speaks of the frequency distribution of (A) well graded soil (B) poorly graded soil and (C) step graded soil ii. Define coefficient of uniformity and coefficient of curvature and discuss their relative significance with reference to soil gradation.	1 4 3 2																										
4	(a) The following state of strain exists at a point 'P'. $[\epsilon_{ij}] = \begin{bmatrix} 270 & 400 & 90 \\ 400 & 680 & 215 \\ 90 & 215 & 570 \end{bmatrix} \times 10^{-3}$ Given a line AB of direction cosines $n_x = 0.6$ $n_y = 0.8$ $n_z = 0$, determine the strain along the line AB and the volumetric strain at point P. (b) i. Write a short note on the concept of position vector and displacement consistent with the rock mechanics convention. ii. The displacement field of a body is given by $\vec{U} = u(x, y, z) \vec{i} + v(x, y, z) \vec{j} + w(x, y, z) \vec{k}$ $= (x^3 + 5) \vec{i} + (5y + x^2) \vec{j} + (2y^2 + z^2) \vec{k}$ Determine the deformed position of a point originally at $(2, 5, -4)$	4+1= (5) 3 2																										
5	(a) Enlist the general assumptions of classical theory of hydraulic fracturing. (b) In hydraulic fracturing experiment, the fracture initiates along the major principal stress direction. Prove the proposition mathematically using neat and labeled diagram. (c) While conducting hydraulic fracturing test in hard and impermeable rock of tensile strength 6.8 MPa, the breakdown pressure was recorded as 8.2 MPa and the shut-in pressure was recorded as 5.2 MPa. Determine the magnitude of maximum horizontal stress.	2 6 2																										
6	Two major immediate roof rock strata have the following ratings of different parameters of rock mass as shown in the table below. Rock strata of type-1 and type-2 are 0.8 m and 1.2 m thick respectively. <table><tr><th rowspan="2">Sl. No.</th><th rowspan="2">Parameters</th><th colspan="2">Rating</th></tr><tr><th>Rock stratum type-1</th><th>Rock stratum type-2</th></tr><tr><td>1</td><td>Layer thickness</td><td>20</td><td>15</td></tr><tr><td>2</td><td>Structural features</td><td>15</td><td>12</td></tr><tr><td>3</td><td>Weatherability</td><td>10</td><td>14</td></tr><tr><td>4</td><td>Strength</td><td>12</td><td>10</td></tr><tr><td>5</td><td>Ground water seepage</td><td>7</td><td>8</td></tr></table> Given the following additional information - Average value of the density of the two rock types = 2.5 te/m^3 - Gallery width = 4.5 m - Pull out strength of each rock bolt = 12 te - Row spacing of the roof bolts = 2.0 m (a) Calculate the weighted average RMR as per ISM-CMRI geomechanics rock mass classification system (b) Calculate the number of bolts in a row to ensure a factor of safety of 1.5. (c) Calculate the final factor of safety so achieved	Sl. No.	Parameters	Rating		Rock stratum type-1	Rock stratum type-2	1	Layer thickness	20	15	2	Structural features	15	12	3	Weatherability	10	14	4	Strength	12	10	5	Ground water seepage	7	8	1 4 2
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(d)	The parameters of the Q-Rock Mass Classification are given in the following table. Determine the weighted average Q-Value $Q = \frac{RQD}{J_n} \times \frac{J_r}{J_a} \times \frac{J_w}{SRF}$ <table border="1"><thead><tr><th>Parameter of Q</th><th>Poorest Value</th><th>Most Typical Value</th><th>Maximum value</th></tr></thead><tbody><tr><td>RQD</td><td>30</td><td>75 63.33</td><td>85</td></tr><tr><td>J_n</td><td>15</td><td>6 7.66</td><td>2</td></tr><tr><td>J_r</td><td>1.5</td><td>3 2.83</td><td>4</td></tr><tr><td>J_a</td><td>7</td><td>2 3.33</td><td>1</td></tr><tr><td>J_w</td><td>0.65</td><td>1 0.883</td><td>1</td></tr><tr><td>SRF</td><td>8.5</td><td>5.5 17.5</td><td>3.5</td></tr></tbody></table>	Parameter of Q	Poorest Value	Most Typical Value	Maximum value	RQD	30	75 63.33	85	J_n	15	6 7.66	2	J_r	1.5	3 2.83	4	J_a	7	2 3.33	1	J_w	0.65	1 0.883	1	SRF	8.5	5.5 17.5	3.5	3
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7	A circular tunnel of 6 m diameter is to be constructed by drilling and blasting (the disturbance factor due to blasting is assumed to be 0.90) at a depth of 300m the long axis of the tunnel is proposed to be aligned along $N20^0$. - The geotechnical consultant has provided the results of in situ stress given below in the table. <table border="1"><thead><tr><th>In situ stress component</th><th>Value</th></tr></thead><tbody><tr><td>Vertical stress, σ_v (MPa)</td><td>8.1</td></tr><tr><td>Major Principal Stress σ_H (MPa)</td><td>20.25</td></tr><tr><td>Minor principal stress σ_h (MPa)</td><td>16.20</td></tr><tr><td>Direction of Major principal stress</td><td>$N20^0$</td></tr></tbody></table> - The intact rock testing in the laboratory revealed the compressive and tensile strength of rock as 63.56 MPa and 2.76 MPa respectively. - The site engineer carried out the detailed geological mapping of an area of $9m^2$ inside an exploratory drift at the proposed tunnel location revealing the following information. - The area consists of 3-joint sets each of an average spacing of 0.6 m, 0.25 m and 0.55 m respectively. In addition to the said three joint sets the area also encountered 15 random joints. - All the joint surfaces are slightly rough having slightly weathered surface containing soft infilling of size of 2 mm and less. (a) Using the concept of Mohr's Circle, determine the factor of safety of the point "P" shown in the figure at a distance of $r = 6\sqrt{2}$. Given that the vector OP (lying in the vertical plane) is inclined at an angle of $\theta = 45^0$ with respect to the horizontal 	In situ stress component	Value	Vertical stress, σ_v (MPa)	8.1	Major Principal Stress σ_H (MPa)	20.25	Minor principal stress σ_h (MPa)	16.20	Direction of Major principal stress	$N20^0$	10																		
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8	Write short notes on any TWO of the followings - Importance of instrumentation in strata monitoring - Typical problems, Critical parameters, Analysis methods and Acceptability Criteria for design of pillars in room and pillar mines - Cassagrande apparatus and its applicability - Construction of Doorstopper cell for in situ stress measurement	2×5=10																												

Department of Mining Engineering

IIT(ISM), Dhanbad

IV Semester B. Tech (Mining Engineering)

End Semester Examination Winter Semester Session 2023-24

Subject: Underground Coal Mining (MNC 208)

Time – 3 hours

Full marks - 100

Attempt any five questions

Question No.	Questions	Marks
1.	a) With the help of neat labelled sketch of a retreating longwall mine producing 10000 tonne coal per day from one working longwall panel and one longwall panel being developed, showing the layout of the following longwall panels with three entry panel system and the inter-panel barrier and barrier between the longwall panel and trunk roads and also showing the ventilation status, quantity of air in each gate road and ventilation control devices like ventilation stoppings, regulators, isolation stoppings etc of the following panels: I. One longwall panel of 3 km long and 300 m wide at retreat of 300 m from the installation chamber II. One extracted longwall panel adjacent to panel being extracted III. One developed longwall panel of 3 km long and 300 m wide adjacent to the panel under extraction, and IV. Three headings of a longwall panel being developed adjacent to the developed panel b) Show the list of major equipment in the mine for working longwall panel and longwall panel being developed c) Calculate the cycle time of the longwall operating face of 300 m wide, assuming your own data as required. You may make your own assumptions, as and if necessary.	12+4+4
2.	Explain why (Any four) a) Caving method is not permitted for extraction of coal seams with high inherent in-situ gas content and working under massive sand stone roof. b) Rider is provided in sinking shaft winding system c) Two legged shield supports are preferred in case of weak immediate roof d) Operation of longwall panels should not be stopped during the periodic weighting	5 X 4

Instruction: Answer ALL questions

Q. 1	a) Write down the objectives of mine ventilation. Or Write down the standards of ventilation as per CMR-2017. b) In a mine ventilation circuit, there are two ventilation splits 'A' and 'B'. Split 'A' draws double the air quantity drawn by split 'B'. Resistance of split 'B' is exactly same as the resistance of the shafts and trunk airways. A regulator is put in split 'A' to make the quantity of air flowing in both the splits equal. The main mine fan produces a pressure of 1000 Pa and supplies 100 m³/s of air in the mine. Calculate the size of the regulator. Assume the fan pressure remains same before and after installation of the regulator.	6 14
Q. 2	A forward-bladed centrifugal fan is installed in a mine having two vertical shafts, each of 6 m diameter, connected at the bottom by a level gallery of 3 m × 4 m cross-section and 800 m long. The fan has an impeller of diameter of 1500 mm, width of 1000 mm and vane angle of 1.2 radians at the outlet. The fan rotates at a speed of 200 rpm and circulates 25 m³/s of air. The coefficient of friction K for the shafts is 0.004 Ns²m⁻⁴ and for the level is 0.01 Ns²m⁻⁴. Neglect the shock loss. With these information, calculate the following: - Theoretical head and air power developed by the fan - Velocity of airflow in the level gallery Assume any other data, if required. <i>ds = 500m</i>	10+10
Q. 3	a) Discuss the sources of generation, physiological effects, and control of Radon gas in underground mine atmosphere. b) With the help of pressure-enthalpy diagram superimposed by the state diagram of the refrigerant, explain the working of vapour-compression refrigeration system used for mine air-conditioning.	8 12
Q. 4	a) With neat sketches discuss any two important ventilation control devices used in underground coal mines. b) What do you mean by N.V.P. and motive column? Calculate the N.V.P. and height of motive column in an underground mine using the following data: Depth of mine = 300 m Pit-bottom barometer reading = 101.27 kPa Pit-top barometer reading = 98.10 kPa Average temperature in downcast shaft = 304 K Average temperature in upcast shaft = 307 K Or A 4.5 m diameter shaft circulating 45 m³/s of air absorbs a pressure of 550 Pa. Find the flow in the shaft when its diameter is enlarged to 6 m assuming the pressure drop across it to remain unchanged. Also, find the pressure drop in the enlarged shaft if the airflow is maintained at 45 m³/s.	4+4 4+8
Q. 5	a) Discuss the applications of booster fan and auxiliary fan. b) Drawing the sketches of both centrifugal and axial-flow fans, compare their constructional features, working principles, characteristics, and merits and limitations.	3+3 4+10

Quiz-02
GPE201 (Earth and Planetary System)

Date: 10 April 2024

Time: 19:15 – 19:45 hrs

Maximum marks: 10

General instructions: There are ten questions in total, and the candidate must attempt all of them. Each question is worth one mark.

Unless otherwise mentioned, symbols and notations have their usual standard meaning. Assume suitable data, if necessary, and indicate the same clearly.

Constants that may be required:

Planck constant: $6.62607015 \times 10^{-34}$ J.s

$B = 1.28 \times 10^{-11} \text{ W m}^{-2} \mu\text{m}^{-1} \text{ K}^{-5}$ where λ_{max} is in μm .

Wien's displacement constant: $\sigma_W = 2.898 \times 10^{-3} \text{ m K}$.

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- Q.1:** What does the term "climate feedback" mean? Provide an example of both a positive and a negative feedback in the climate system.
- Q.2:** What are three key factors, besides insolation and greenhouse gases, that have played a significant role in altering Earth's climate over geological time?
- Q.3:** Determine the temperature in degrees Celsius of a star that emits light at a peak intensity wavelength of 400 nanometers.
- Q.4:** Provide a neat and labeled diagram that clearly shows the vertical temperature distribution in the ocean during the summer and winter seasons.
- Q.5:** What does climate sensitivity mean, and what is the most widely accepted estimate for it?
- Q.6:** What is the Rayleigh-Jeans catastrophe, and how did it contribute to the development of quantum mechanics?
- Q.7:** What are the four primary gases in Earth's atmosphere, and what are their relative proportions?
- Q.8:** What does AOD mean in atmospheric science? What are the typical values of AOD for extremely clean and very hazy atmospheric conditions?
- Q.9:** Enumerate at least eight factors that influence the distribution of temperature in ocean water.
- Q.10:** What is the maximum monochromatic exitance of isotherm of a blackbody at a temperature of 2150 K, with a peak wavelength of $1.219 \mu\text{m}$.
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Answer any five Questions.

Maximum marks: 100

Time: 3:00hrs

SL. No.	Description of Question	Marks
1.0	a) Briefly Discuss on the following: i. Factors influencing development requirements ii. Different Purposes served by the mine development iii. Excavation Methods and its advantages & Disadvantages	10
	b) Briefly discuss the conventional method of direction and gradient control of a drift along with working principle of automated guidance system for direction and gradient control used in TBM.	10
2.0	A drift of 100 m length and opening size 4.5m x 3.2 m is to be driven from Point "A" to "B" in an underground metal mine of uranium in India using drilling and blasting method of excavation. Drift is to be developed using trackless mining system with single boom drill jumbo of 3.2 m drill rod, LHD and LPDT combination of 5 tonne and 15 tonne capacity, respectively, etc. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Steps involved in the development of the Drift ii. Manpower distribution plan activity wise/shift-wise iii. Days required to complete the development of the drift iv. Equipment/Machineries deployed/needed v. Drilling and blasting pattern vi. Drilling meterage per round of blasting vii. Explosive consumption per round of blasting viii. Tonnage produced per round of blasting ix. Powder factor obtained x. Ventilation circuit for ventilating the face	8 2 2 1 2 1 1 1 1 1
3.0	A stope raise of an opening size 2m x 2m is to be developed in between two levels of vertical interval 45m in an underground metal mine of uranium using conventional raising method. The dip of the ore body is 30°. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Calculate the length of the Stope Raise and S/R X-cut, if raise is developed at 42° ii. Steps involved in the raising process using Conventional Raising iii. Manpower distribution plan activity wise/shift-wise iv. Days required to complete the raise v. Equipment/Machineries deployed/needed vi. Drilling and blasting pattern vii. Drilling meterage per round of blasting viii. Explosive consumption per round of blasting ix. Tonnage produced per round of blasting x. Powder factor obtained xi. Ventilation circuit for ventilating the face	1 7 2 2 1 2 1 1 1 1 1

4.0	a) Why raising is preferred over winzing? Briefly discuss the different raising methods.	5
	b) A Horizontal Cut and Fill stoping block of length 200m in between two levels (level interval is 45m) is to be developed in an underground metal mine of uranium with trackless mining system with single boom drill jumbo of 3.2 m drill rod, LHD and LPDT combination of 5 tonne and 15 tonne capacity, respectively, etc. Dip of the ore body is 30° and plan thickness is 9 m. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Type and size of the excavations, required for its development, in sequence ii. Days required to complete the stoping block development	5 5
	c) Tabulate the applicable stoping method(s) based on the geological and geo-technical aspects.	5
5.0	a) Briefly compare the advantages and disadvantages of different stoping methods	10
	b) Briefly discuss the Systematic Support Rules (SSR) under Regulation 112 of MMR 1961 for a typical trackless underground metal mine in India (adopted horizontal cut and fill stoping method) applicable to all workings of mine where RMR of Back/Hanging Wall/Ore body varies from 65 to 80.	10
6.0	A developed stoping block of 200m for Horizontal Cut and Fill (HCF) in between two levels (level interval is 45m) is to be stoped-out with trackless mining system with single boom drill jumbo of 3.2 m drill rod, LHD and LPDT combination of 5 tonne and 20 tonne capacity, respectively, etc. Dip of the ore body is 30° and plan thickness is 9m. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Steps involved in the stoping process ii. Manpower distribution plan activity wise/shift-wise iii. Days required to stoped-out developed block completely iv. Equipment/Machineries deployed/needed v. Different Drilling and blasting patterns adopted while stoping process starts vi. Drilling meterage per round of blasting in different cases vii. Explosive consumption per round of blasting in different cases viii. Tonnage produced per round of blasting in different cases ix. Powder factor obtained in different cases x. Ventilation circuit for ventilating the stoping area	7 2 2 1 3 1 1 1 1 1

Quiz-01
GPE201 (Earth and Planetary System)

Date: 09 February 2024

Time: 19:00 – 19:30 hrs

Maximum marks: 10

General instructions: There are a total of ten questions, and the candidate must attempt all of them. Each question is worth one mark.

Unless otherwise mentioned, symbols and notations have their usual standard meaning. Assume suitable data, if necessary, and indicate the same clearly.

Q.1: How does the seismic reflection method differ from the magnetic method?

Q.2: Using the concepts of flattening (f) and eccentricity (e) of the Earth, demonstrate that $e^2 = f(2-f)$.

Q.3: What does the term "Great Oxidation Event" refer to?

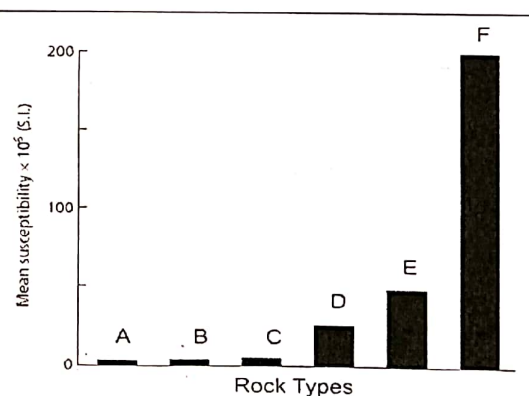
Q.4: Determine the mass of the Sun, given the orbital radius of the Earth as $1.496 \times 10^{11} \text{m}$ and the universal gravitational constant (G) as $6.67 \times 10^{-11} \text{Nm}^2/\text{kg}$.

Q.5: A rock sample, with a magnetic susceptibility of 2.8×10^{-3} , is magnetized in a magnetic field intensity of $0.5 \times 10^4 \text{ A/m}$. What will be the magnetization of the rock sample?

Q.6: What are Perihelion and Aphelion? Compute the distance between the Sun and the Perihelion and Aphelion in terms of eccentricity and the semi-major and minor axes of the orbital pathway.

Q.7:

The adjacent diagram illustrates the magnetic susceptibility of typical rock types. Identify rock types A through F.



Q.8: Explain the mechanism accountable for the formation of the elements found within the periodic table.

Q.9: What does the term "Eon" signify within the geological time scale?

Q.10: A seismic survey was conducted in a region with two geological layers: the upper layer comprises shale with a density of 2300 kg/m^3 and a seismic velocity of 2950 m/s , while the lower layer is composed of sandstone with a density of 2200 kg/m^3 and a seismic velocity of 2400 m/s . Calculate the reflection coefficient and explain its significance.

Mid-Semester Examination (WS: 2023-24)
GPE201 (Earth and Planetary System)

Date: 27 February 2024

Time: 14:00 – 16:00 hr

Maximum marks: 32

General instructions: There are Ten questions in total, and the candidate must attempt any eight of them. Parts of a question must be answered sequentially in the same place. The number of marks assigned to each question/part is indicated next to it. Unless otherwise mentioned, symbols and notations have their usual standard meaning. Assume suitable data if necessary, and indicate the same clearly.

Constants that may be required:

Avogadro's number: $6.02214076 \times 10^{23} \text{ mol}^{-1}$

Average distance between the Sun and Earth: 150 million kilometers

Planck constant: $6.62607015 \times 10^{-34} \text{ J}\cdot\text{s}$

-
- Q.1: How and when did planet Earth get its oxygen?..... (3+1)
- Q.2: Explain the standards set by the International Astronomical Union for categorizing objects as dwarf planets. Offer illustrations of recognized dwarf planets and elucidate how their attributes distinguish them from typical planets..... (2+1+1)
- Q.3: Discuss the origin, extent, and relief of the mid-ocean ridges. Why is there less sediment on the seafloor near the mid-ocean ridges than elsewhere?..... (1+1+1+1)
- Q.4: Calculate the circumference of the Sun, given that it subtends an angle of approximately 30 arcminutes when observed from the Earth, and its average distance from Earth is $1.50 \times 10^8 \text{ km}$(4)
- Q.5: Describe the Wilson Cycle, emphasising the motion involved in each stages and the physiography associated with it..... (4)
- Q.6: Identify at least four evidences used by Alfred Wegener to support his idea of moving continents. What was the primary technical weakness in Alfred Wegener's theory of continental drift?.....(3+1)
- Q.7: Describe the processes that lead to Moonquakes and Marsquakes. Are these seismic activities caused by traditional plate tectonics on these celestial bodies?.....(3+1)
- Q.8: Provide a concise overview of how minerals demonstrate their properties through color, streak, and luster.....(4)
- Q.9: The radioactive isotope ^{14}C has a half-life of 5730 years. If a system starts with 10^{-11} g of ^{14}C at time $t = 0$, how much ^{14}C remains after 2000 years and 1 million years? What is the significance of these values?(4)
- Q.10: What is Bode's law about solar systems? Calculate the distance between the Sun and Jupiter using the Bode's law..... (2+2)
-

$$\frac{\Delta \theta}{\Delta t} = \tan \theta$$

$$\theta = \tan^{-1}\left(\frac{\Delta \theta}{\Delta t}\right)$$

IV Semester B. Tech (Mining Engineering)

Mid-Semester Examination (Winter Session)

Session: 2023-24

Subject: Underground Coal Mining (MNC 208)

Time – 2 hours

Total marks - 70

Question No. 1 & 7 are compulsory and answer any three from the rest.

Qn. No.	Questions	Marks
1.	(a) With the help of a neat sketch explain how a coal pillar of 30m x 30m centre to centre distance and gallery size of 4m x 2.5m is extracted using split and slicing method of mining. (b) Also calculate the % extraction of the pillar by this method. Seam thickness is 4m. Assume your own condition as and if necessary.	10+10
2.	Explain in brief the advantages and disadvantages of 'Whole' followed by 'Broken' system of Bord and Pillar mining	12
3.	Compare between Bord & Pillar and Longwall method of mining.	12
4.	(a) Explain in brief how location of shafts are selected with respect to deposit's geometry and surface topography. (b) Explain the function of scaffold in a sinking shaft.	7+5
5.	(a) Explain in brief the straight line and diagonal line of extraction. (b) What do you understand by goaf edge support and how goaf edge supports are provided in a bord & pillar panel?	8+4
6.	(a) Explain in brief three major hazards of Underground coal mining. (b) Do you think the concept of determining the size of a panel based on incubation period is justifiable? If not, why?	6+6
7.	(a) Draw a simple layout of a of 5 headings bord & pillar development panel with gallery size of 4m x 2.5m, showing the ventilation and support system. (b) Provide a list of equipment to be used for producing 300 tonne per day from the development panel. (c) How much coal may be produced by each round of blast and how many rounds of blasting are required for producing 300 TPD?	7+3+4

Answer All the Questions.

Maximum marks: 60

Time: 2:00hrs

Sl. No.	Description of Question	Marks
1.0	Summarize the operational underground metal mines in India company wise with salient features in tabular form.	10
2.0	Characterize the primary and secondary development of underground metal mine based on the Inclination, Location, shape, size, and purposes.	15
3.0	Briefly Discuss on the following: a) Factors influencing development requirements b) Different Purposes served by the mine development c) Excavation Methods and its advantages & Disadvantages	10
4.0	A drift of 100 m length and opening size 4.5m x 3.2 m is to be driven from Point "A" to "B" in an underground metal mine using drilling and blasting method of excavation. Drift is to be developed using trackless mining system with single boom drill jumbo of 3.2 m drill rod, LHD and LPDT combination of 5 tonne and 15 tonne capacity, respectively, etc. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed: i. Steps involved in the development of the Drift ii. Days required to complete the development of the drift iii. Equipment/Machineries deployed/needed iv. Drilling and blasting pattern v. Drilling meterage per round of blasting vi. Explosive consumption per round of blasting vii. Tonnage produced per round of blasting viii. Powder factor obtained ix. Ventilation circuit for ventilating the face	10 2 2 3 2 2 1 1 2

Department of Mining Engineering

MNC 208 IV SEM (MNE) 2023-24 QUIZ 1 Date: 21.02.2024

Full marks – 10

Time – 30 minutes

Q1. Explain how choice of mining methods is affected by 5

- (a) Total reserves available in the mine
- (b) Structure and geological anomalies

Q 2. Explain why ventilation of UG workings is not efficient in B&P System compared to Longwall system of mining? 2

Q. 3. Explain in brief three hazards of shaft sinking 3



Instructions: 1. Answer ANY TWO questions

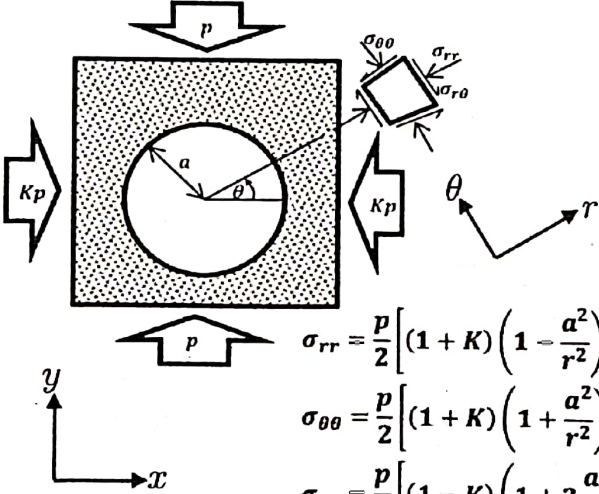
Time: 2 HOURS

2. Neat and accurate sketches will assist you in solving the problems logically and faster

3. Assume logical data, where necessary

Max. Marks: 30

4. Use the sign conventions applicable for rock mechanics.

Q. No.	Question	Marks
1	(i) Draw neat sketches and discuss the Cauchy's First and Second law of stress. (ii) Draw the Mohr's Circle for (a) a rock core specimen loaded in uni-axial compression (b) a thin square rock slab loaded in tension from two of its side. (c) A square rock slab loaded hydro-statically (iii) Kirsch in 1898, considered a circular opening in an infinite medium subjected to horizontal stress of $K \times p$ and a vertical stress of p. These stresses are known as the boundary stress or in situ stress. He gave the solution for finding out the components of stress tensor (σ_{rr}, $\sigma_{\theta\theta}$ and $\sigma_{r\theta}$) around a circular opening in the polar co-ordinate system with its origin at the center of the circular excavation, shown in the figure below. (a) Using the formula given below, determine the components of the stress tensor at point 'M' in the direction of 45° to the horizontal at an infinite distance. (b) At the said point 'M', determine the components of the stress tensor in x-y coordinate system. (c) Explain the inferences you draw from the results of point (a) and point (b)  $\sigma_{rr} = \frac{p}{2} \left[(1+K) \left(1 - \frac{a^2}{r^2} \right) - (1-K) \left(1 - 4 \frac{a^2}{r^2} + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$ $\sigma_{\theta\theta} = \frac{p}{2} \left[(1+K) \left(1 + \frac{a^2}{r^2} \right) + (1-K) \left(1 + 3 \frac{a^4}{r^4} \right) \cos 2\theta \right]$ $\sigma_{r\theta} = \frac{p}{2} \left[(1-K) \left(1 + 2 \frac{a^2}{r^2} - 3 \frac{a^4}{r^4} \right) \sin 2\theta \right]$	6 1 1 1 1 4 1
2	(i) The rectangular state of stress at a point 'P' in a rock is given by the array $\begin{bmatrix} 5 & 5 & 3 \\ 5 & -5 & 4 \\ 3 & 4 & 8 \end{bmatrix}$ Determine the magnitude of the normal stress and the shear stress on a given plane. The direction cosines of the unit normal to the given plane are $n_x = \sqrt{\frac{1}{3}}$ and $n_y = \sqrt{\frac{1}{2}}$.	5

Class Test 2
IV Sem (Mining Engineering)
Underground Coal Mining (MNC 504)
Winter Semester
Session 2023-24

Qn. No.	Question	Total Marks - 10
1	Explain in brief the functions of a) Setting valve b) Positive Set Valve c) Guaranteed Set Valve d) Yield Valve	Marks 2+1+1+3
2	With the help of suitable sketch explain what you understand by periodic weighting in a longwall panel.	3